The Fractional Riccati Equation, Bernstein–Widder, and Fox–H: A Complete Exposition

The paper `frmp.tex` constructs a closed-form representation of the rough Heston density and European call price as finite sums of Fox H-functions, built from a Müntz–Tau series for the fractional Riccati solution, resummed by Padé approximation, and inverted via Mellin–Barnes. This exposition explains how every layer of that construction is the constructive image of one foundational theorem — Bernstein's characterization of completely monotone functions, as sharpened by Widder — applied successively to the kernel of the Riemann–Liouville integral, to the Mittag-Leffler functions that solve the linearized fractional equation, and finally to the Fox H representation of the density itself.

Nothing in what follows is loose analogy: each completely monotone object that appears in the rough Heston construction is the Laplace transform of a positive measure, and that measure is realized concretely in the paper.

1. The Object: The Fractional Riccati Equation in `frmp.tex`

The central nonlinear equation of the paper (equation (3.3) in the manuscript, written `eq:heston_IVP`) is the Volterra integral equation

$$h(v,t) = I^{\mu}\big[\, P(v) + Q(v)\, h(v,t) + R\, h(v,t)^2 \,\big], \qquad h(v,0) = 0,$$

where the coefficients are quadratic polynomials in the Fourier variable v:

$$P(v) = \tfrac{1}{2}(-v^2 - iv), \qquad Q(v) = \lambda(iv\rho\nu - 1), \qquad R = \tfrac{1}{2}\lambda^2\nu^2,$$

and I^{μ} is the Riemann–Liouville fractional integral of order $\mu = H + \frac{1}{2} \in (\frac{1}{2}, 1)$, with kernel $(t-s)^{\mu-1}/\Gamma(\mu)$ (Definition 1.1 in the paper):

$$I^{\mu} f(t) = \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} f(s)\, ds. \tag{1}$$

The function $h(v,\cdot)$ is the fractional analogue of what, in the classical Heston model ($\mu = 1$), would be the solution of the matrix Riccati ODE. The role of h in pricing is to assemble the cumulant generating function

$$\Phi(v,T) = \theta\lambda \int_0^T h(v,s)\, ds + V_0\, I^{1-\mu} h(v,\cdot)(T),$$

so that the characteristic function of the log-return is $\phi_T(v) = e^{\Phi(v,T)}$ (the affine-exponential structure inherited from El Euch–Rosenbaum 2019).

The two structural facts that drive everything else are:

(a) The kernel of I^μ, namely $t^{\mu-1}/\Gamma(\mu)$ for $0 < \mu < 1$, is **completely monotone** on $(0, \infty)$.

(b) When the quadratic term in (1) is switched off (i.e.\ $R = 0$), the equation reduces to a fractional linear equation whose solutions are **Mittag-Leffler functions**, which are themselves completely monotone for real negative argument.

Both of these objects, by Bernstein's theorem, are Laplace transforms of positive measures. The paper's Müntz–Padé–Fox-H machinery is, structurally, the explicit construction of those measures.

2. Bernstein's Theorem and the Widder Sharpening

A function $f : (0, \infty) \to [0, \infty)$ is **completely monotone** if it is C^∞ and

$$(-1)^n f^{(n)}(t) \geq 0 \qquad \text{for every } n \geq 0,\ t > 0.$$

Bernstein's theorem (1928). A function f is completely monotone on $(0, \infty)$ if and only if there exists a unique nonnegative Borel measure μ_f on $[0, \infty)$ such that

$$f(t) = \int_0^\infty e^{-st}\, d\mu_f(s), \qquad t > 0. \tag{2}$$

Widder's sharpening (1941). The measure μ_f is recovered from f by the post-Widder inversion formula

$$d\mu_f(s) = \lim_{n\to\infty} \frac{(-1)^n}{n!}\Big(\frac{n}{s}\Big)^{n+1} f^{(n)}\Big(\frac{n}{s}\Big)\, ds,$$

so the map $f \mapsto \mu_f$ is a bijection from completely monotone functions onto positive measures with all moments finite enough to support (2). The pair (Bernstein existence, Widder inversion) we will refer to collectively as **Bernstein–Widder**.

Two structural consequences are used repeatedly:

(B1) **Closure under products.** If f_1, f_2 are completely monotone with measures μ_1, μ_2, then $f_1 f_2$ is completely monotone with measure $\mu_1 * \mu_2$ (convolution on $[0, \infty)$).

(B2) **Closure under composition with Bernstein functions.** If ψ is a Bernstein function (i.e.\ ψ' is completely monotone, $\psi(0) \geq 0$), and f is completely monotone, then $f \circ \psi$ is completely monotone.

These two closure properties, applied to the rough Heston construction, are what allow the **finite-convolution structure** of the Fox-H density (Theorem 5.1 in the paper, equation `eq:foxH_density`) to be derived purely from Bernstein–Widder algebra on the partial-fraction factors of ϕ_M.

3. The Riemann–Liouville Kernel Is Completely Monotone

For $0 < \mu < 1$ the kernel of I^μ is $K_\mu(t) = t^{\mu-1}/\Gamma(\mu)$. One verifies directly:

$$(-1)^n \frac{d^n}{dt^n}\, t^{\mu-1} = (1-\mu)(2-\mu)\cdots(n-\mu)\, t^{\mu-1-n} \geq 0 \quad (t>0),$$

since each factor $(k-\mu) > 0$ for $k \geq 1$ when $\mu < 1$. So K_μ is completely monotone. Its Bernstein–Widder representation is explicit: from $\mathcal{L}\{t^{\mu-1}\}(s) = \Gamma(\mu)/s^\mu$, inverting Laplace gives

$$\frac{t^{\mu-1}}{\Gamma(\mu)} = \int_0^\infty e^{-st}\, \frac{s^{-\mu}}{\Gamma(1-\mu)}\, ds. \tag{3}$$

The Bernstein measure of the kernel is

$$d\mu_{K_\mu}(s) = \frac{s^{-\mu}}{\Gamma(1-\mu)}\, ds,$$

which is exactly the **stable subordinator's Lévy density** at level μ (up to a normalizing constant) — the same one that produces μ-stable subordination of Brownian motion to give fractional Brownian–type behavior. The connection is not decorative: the Riemann–Liouville operator I^μ is the convolution with the kernel that is the marginal of a μ-stable subordinator, and that subordinator is precisely the process whose passage-time density underlies the rough Heston "fractional time change" picture.

What this means operationally in `frmp.tex`. The fact that K_μ is a positive convolution against a positive measure is what makes I^μ a positivity-preserving bounded operator from $L^1(0,T)$ to $L^1(0,T)$. When the paper asserts in Lemma 1.2 (the power rule $I^\mu t^s = \Gamma(s+1)/\Gamma(s+\mu+1)\, t^{s+\mu}$) that the Müntz lattice $\{t^{k\mu}\}_{k\geq 0}$ is closed under I^μ and under multiplication, the closure under multiplication is trivial and the closure under I^μ is exactly what the Beta-function identity gives. But the **boundedness in** $C[0,T]$ of the iteration of I^μ that underlies the convergence of the Müntz–Tau recurrence (Theorem 2.1) is a consequence of (3): one is iterating a Laplace transform of a positive measure, and contraction follows from a Gronwall–type estimate that uses positivity in a crucial way.

4. The Mittag-Leffler Function as the Eigenfunction of D^μ, and Its Complete Monotonicity

Strip the quadratic term in (1) by setting $R = 0$ and take $P(v) = c_0$, $Q(v) = c_1 = \lambda$. The equation reduces to the **fractional linear Volterra equation**

$$y(t) = I^\mu\big[c_0 + \lambda\, y(t)\big], \qquad y(0) = 0.$$

Its closed-form solution is

$$y(t) = \frac{c_0}{\lambda}\,\big[E_\mu(\lambda t^\mu) - 1\big],$$

where

$$E_\mu(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\mu k + 1)}$$

is the **one-parameter Mittag-Leffler function.** The reason this works is that $E_\mu(\lambda t^\mu)$ is the eigenfunction of the Caputo derivative D_C^μ (and equivalently the resolvent of I^μ):

$$D_C^\mu\, E_\mu(\lambda t^\mu) = \lambda\, E_\mu(\lambda t^\mu).$$

In the classical limit $\mu \to 1$, $E_1(z) = e^z$, and one recovers the ordinary exponential eigenfunction of d/dt. The Mittag-Leffler function is therefore the **exact fractional analogue of the exponential**, and it is the natural building block of any linear theory of I^μ.

The Pollard Theorem (1948)

The decisive analytic fact is that for $0 < \mu \le 1$, the function

$$t \;\longmapsto\; E_\mu(-t^\mu)$$

is **completely monotone** on $[0, \infty)$. This is Pollard's theorem. By Bernstein–Widder, there exists a unique positive measure $\mathfrak{m}_\mu$ on $[0, \infty)$ with

$$E_\mu(-t^\mu) = \int_0^\infty e^{-st}\, d\mathfrak{m}_\mu(s). \tag{4}$$

Pollard exhibits the measure explicitly. Define the **M-Wright function** (sometimes called the Mainardi function or Wright function of the second kind)

$$M_\mu(r) := \sum_{n=0}^{\infty} \frac{(-r)^n}{n!\,\Gamma(-\mu n + 1 - \mu)} = W_{-\mu,\,1-\mu}(-r),$$

where $W_{\alpha,\beta}(z) = \sum_{n\ge 0} z^n/[n!\,\Gamma(\alpha n + \beta)]$ is the generic Wright function. Pollard's identity is

$$E_\mu(-t^\mu) = \int_0^\infty e^{-tr}\, M_\mu(r)\, dr, \qquad t \ge 0. \tag{5}$$

So $d\mathfrak{m}_\mu(r) = M_\mu(r)\, dr$ — the M-Wright density is the Bernstein–Widder spectral density of the Mittag-Leffler function $E_\mu(-t^\mu)$. It is positive, integrates to one (since $E_\mu(0) = 1$), and is the density of the **inverse μ-stable subordinator** at time $t = 1$.

The Two-Parameter Family $E_{\mu,\mu}(-\lambda t^\mu)$

The paper's coefficient $Q(v) = \lambda(iv\rho\nu - 1)$ introduces a mean-reversion rate $\lambda > 0$, so the relevant Mittag-Leffler factor is the **two-parameter** one,

$$E_{\mu,\beta}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\mu k + \beta)},$$

evaluated at $z = -\lambda t^\mu$ with $\beta = \mu$. The Schneider theorem (1996) extends Pollard: $t \mapsto E_{\mu,\mu}(-\lambda t^\mu)$ is completely monotone for every $0 < \mu \le 1$, $\lambda \ge 0$, with Bernstein–Widder representation

$$E_{\mu,\mu}(-\lambda t^\mu) = \int_0^\infty e^{-st}\, p_{\mu,\lambda}(s)\, ds, \tag{6}$$

where the spectral density $p_{\mu,\lambda}$ is

$$p_{\mu,\lambda}(s) = \frac{1}{\pi}\, \frac{\lambda\, s^{\mu-1} \sin(\mu\pi)}{s^{2\mu} + 2\lambda s^\mu \cos(\mu\pi) + \lambda^2}, \qquad s > 0.$$

This is the spectral function that the paper implicitly inverts — every partial-fraction pole of Φ_M in the variable v carries, upon Fourier inversion in the conjugate variable x, exactly such a Mittag-Leffler-type density. The Fox-H closed form of Theorem 5.1 is the explicit name for these densities once they are written through their Mellin–Barnes integrals.

Why $E_{\mu,\mu}(-\lambda t^\mu)$ and Not $E_\mu(-\lambda t^\mu)$?

The Mellin–Barnes representation of the M-Wright function is

$$M_\mu(r) = \frac{1}{2\pi i} \int_{\mathcal{L}} \frac{\Gamma(s)}{\Gamma(1-\mu+\mu s)}\, r^{-s}\, ds,$$

which is Fox H with parameters $(m, n, p, q) = (1, 0, 1, 1)$ and data $(a_1, \alpha_1) = (0, \mu)$, $(b_1, \beta_1) = (0, 1)$. More generally, every Mittag-Leffler-like density that arises in (6) admits a Mellin–Barnes form, hence is a Fox H. This is the **point of contact** between the fractional-calculus side (Mittag-Leffler) and the special-function side (Fox H) of the paper.

5. The Prabhakar Extension and Its Bernstein Measure

The full nonlinear Riccati equation in (1), with $c_2 = R \neq 0$, does not solve in closed form. But the **structure of its perturbation expansion** is the central object of the paper's Müntz–Tau recurrence (Theorem 2.1):

$$h(v, t) = \sum_{k \ge 1} a_k(v)\, t^{k\mu},$$

with

$$a_1 = \frac{c_0}{\Gamma(\mu+1)}, \qquad a_k = \gamma_k \left[c_1 a_{k-1} + c_2 \sum_{j=1}^{k-2} a_j\, a_{k-1-j} \right], \quad k \ge 2,$$

and $\gamma_k = \Gamma((k-1)\mu + 1)/\Gamma(k\mu + 1)$. Each term $t^{k\mu}$ in this expansion is, by (3) applied iteratively, expressible as a Laplace transform of a Lévy-stable-like density. When the series is summed formally (and Padé-resummed to extend convergence beyond the Puiseux radius — Theorem 4.1, Stahl convergence), the result is a rational function in $z = T^\mu$ whose poles produce, upon Fourier inversion, **Prabhakar-type Mittag-Leffler factors**:

$$E^{\gamma}_{\mu,\beta}(z) = \sum_{k=0}^{\infty} \frac{(\gamma)_k\, z^k}{k!\,\Gamma(\mu k + \beta)},$$

with $(\gamma)_k$ the Pochhammer symbol. Mainardi–Garrappa (2014) proved that for the parameter range $0 < \mu \le 1, 0 < \beta \le 1, 0 < \gamma \le \beta/\mu$, the function $t \mapsto t^{\beta-1} E^{\gamma}_{\mu,\beta}(-\lambda t^{\mu})$ is completely monotone, with Bernstein–Widder density expressible through the Stieltjes transform of an explicit nonnegative function.

The relevance to `frmp.tex`: the **Padé partial-fraction expansion** of Φ_M (equation `eq:PFE`),

$$\Phi_M(u,T) = -\tfrac{1}{2}\sigma_T^2 u^2 - i\mu_T u + \sum_{j=1}^{2M} \frac{A_j(T)}{u - u_j(T)},$$

exponentiates to

$$\phi_M(u,T) = e^{-\frac{1}{2}\sigma_T^2 u^2 - i\mu_T u} \cdot \prod_{j=1}^{2M} \exp\left(\frac{A_j(T)}{u - u_j(T)}\right),$$

and each pole factor $\exp(A_j/(u-u_j))$ is, in the variable conjugate to u via Fourier inversion, **a Mittag-Leffler-type completely monotone factor** (in fact, a modified Bessel / Wright-function density). The Bernstein measure of each such factor is exactly what appears in the corresponding Fox-H summand of Theorem 5.1.

6. From the Pole Factor to the Fox-H Summand

This is the technical heart of the connection. Take a single pole factor

$$g_{A,u_0}(u) := \exp\left(\frac{A}{u - u_0}\right), \qquad u_0 \notin \mathbb{R},\ A \in \mathbb{C}.$$

Set . The Fourier inverse of g_{A,u_0}, in the variable x conjugate to u, is computed in three steps.

Step 1: Series expansion. Expand the exponential:

$$g_{A,u_0}(u) = \sum_{n=0}^{\infty} \frac{A^n}{n!\,(u-u_0)^n}.$$

Step 2: Fourier inverse term by term. The Fourier inverse of $1/(u-u_0)^n$ is, by residue calculus (closing in the half-plane opposite to u_0),

$$\mathcal{F}^{-1}\left[\frac{1}{(u-u_0)^n}\right](x) = \frac{(-i)^n\,\epsilon}{(n-1)!}\, x^{n-1}\, e^{-iu_0 x}\, \mathbf{1}_{\epsilon x > 0}, \quad n \ge 1.$$

(The $n=0$ term gives a delta, treated separately and absorbed into the trivial $S=\emptyset$ Gaussian summand of Theorem 5.1.)

Step 3: Resum the Bessel series. Substituting and reorganizing,

$$\mathcal{F}^{-1}[g_{A,u_0}](x) = \epsilon\,\delta(x)\cdot[\text{trivial}] + \epsilon\, e^{-iu_0x}\,\mathbf{1}_{\epsilon x>0}\sum_{n=1}^{\infty}\frac{(-iA)^n}{n!\,(n-1)!}\,x^{n-1}.$$

The inner series is the **modified Bessel function** I_0 up to normalization:

$$\sum_{n=1}^{\infty}\frac{(-iAx)^{n-1}\,(-iA)}{n!\,(n-1)!} = -iA\,\frac{I_1(2\sqrt{-iAx})}{\sqrt{-iAx}} = -iA\;\cdot(\text{Bessel}).$$

The Bessel function admits the Mellin–Barnes representation

$$I_\nu(2\sqrt{z}) = \frac{1}{2\pi i}\int_{\mathcal{L}}\frac{\Gamma(-s)}{\Gamma(\nu+1+s)}\,(-z)^s\,ds,$$

which is exactly Fox $H^{1,0}_{0,2}$ with appropriate parameters. The single-pole Fourier inverse is therefore a **single Fox H function** with one Gamma factor in the numerator and two in the denominator — what the paper labels $(m,n,p,q)=(1,0,0,2)$ or, after absorbing the Gaussian envelope and the support indicator, $(m^\circ_S,n^\circ_S,p^\circ_S,q^\circ_S)=(|S|+1,|S|,|S|+1,2|S|+1)$ at $|S|=1$.

Step 4: Multiply over poles. The full characteristic function factors over poles:

$$\phi_M(u,T) = G_T(u)\prod_{j=1}^{2M} g_{A_j,u_j}(u),$$

with $G_T(u)=\exp(-\frac{1}{2}\sigma_T^2u^2 - i\mu_Tu)$ the Gaussian piece. By the Bernstein–Widder closure property (B1) — products of Laplace transforms of positive measures correspond to convolutions of those measures — the inverse Fourier transform of ϕ_M is

$$f_M(x,T) = (G_T * g_{A_1,u_1} * \cdots * g_{A_{2M},u_{2M}})^\vee(x).$$

Expanding each factor's series and exchanging the order of summation gives the **subset sum** of Theorem 5.1:

$$f_M(x,T) = \sum_{S\subseteq\{1,\ldots,2M\}} d_S(T)\,H^{m^\circ_S,n^\circ_S}_{p^\circ_S,q^\circ_S}\left[\eta_S(x,T)\,\middle|\,\cdots\right].$$

Each summand is the Bernstein–Widder spectral density associated with the subset of poles S — concretely, the multiple convolution of the $|S|$ one-pole Mittag-Leffler-type densities against the Gaussian envelope. The Fox-H function names the convolution.

7. The Bromwich–Mellin Loop: Why Inversion Yields a Fox H

The Mellin–Barnes contour integral that defines the Fox H (Definition 5.1 in `frmp.tex`, equation `eq:foxH_def`) is

$$H^{m,n}_{p,q}\left[z\,\middle|\,\cdots\right] = \frac{1}{2\pi i}\int_{\mathcal{L}}\frac{\prod_{j=1}^{m}\Gamma(b_j+\beta_js)\,\prod_{i=1}^{n}\Gamma(1-a_i-\alpha_is)}{\prod_{j=m+1}^{q}\Gamma(1-b_j-\beta_js)\,\prod_{i=n+1}^{p}\Gamma(a_i+\alpha_is)}\,z^{-s}\,ds.$$

This is structurally identical to a Bromwich (inverse Laplace) integral, with the Gamma-function ratio playing the role of the Laplace transform of a positive measure. The reason every Bernstein–Widder spectral density of a Mittag-Leffler factor lands in Fox-H form is the following identification (Mathai–Saxena–Haubold, Theorem 1.5.1):

> The Mellin transform of $t^{c-1}E^{\gamma}_{\mu,\beta}(-\lambda t^{\mu})$ is
> $$\int_0^\infty t^{c-1}\,t^{\beta-1}E^{\gamma}_{\mu,\beta}(-\lambda t^{\mu})\,dt = \frac{\lambda^{-(c+\beta-1)/\mu}}{\mu}\cdot\frac{\Gamma((c+\beta-1)/\mu)\,\Gamma(\gamma-(c+\beta-1}{\Gamma(\gamma)\,\Gamma(\beta-(c+\beta-1)/\mu\cdot\mu\cdots}$$

(The exact Gamma ratio is given by the Mellin–Barnes definition of $E^{\gamma}_{\mu,\beta}$ itself.) The Bromwich inversion of this ratio against z^{-s} reconstructs $E^{\gamma}_{\mu,\beta}$ up to power factors — but expressed as a Fox H. So:

$$t^{\beta-1}E^{\gamma}_{\mu,\beta}(-\lambda t^{\mu}) = H^{1,1}_{1,2}\left[\lambda t^{\mu}\,\middle|\,\begin{matrix}(1-\gamma,\,1)\\(0,\,1),\ (1-\beta,\,\mu)\end{matrix}\right]\cdot[\text{constant}].$$

This is the **identification** that justifies calling the closed form in Theorem 5.1 the Fox-H representation of the rough-Heston density: each completely monotone Mittag-Leffler-type factor in the Bernstein–Widder expansion of ϕ_M is, by this Mellin–Barnes identity, a Fox H. The summation over subsets S in `eq:foxH_density` is the explicit assembly of the convolutions guaranteed abstractly by (B1).

The "two analytically independent routes" of Theorem 5.2 (the convolution chain and the direct Mellin–Barnes inversion via the Bessel resummation `eq:bessel_resum`) are exactly the two ways of looking at this picture:

(a) **Convolution chain**: start from the Bernstein–Widder factorization of ϕ_M over its pole structure, convolve the spectral measures, and identify the result.

(b) **Direct Mellin–Barnes**: start from the Lewis representation `eq:lewis` (a Bromwich contour integral for the price), substitute the rational ϕ_M, and use the Bessel-series resummation identity `eq:bessel_resum` to identify the result as Fox H.

That both routes deliver identical parameter data is the substantive content of Corollary 5.3 — and it is, structurally, the assertion that the Bernstein–Widder spectral measure is *unique* (Widder's part of the theorem).

8. The Padé Resummation as Modal Truncation in the Fourier Variable

The paper's Padé construction (Section 3, via the Chebyshev algorithm of Theorem 3.2) produces, from the Müntz moments $\{M_k(v)\}_{k=0}^{2M-1}$, a diagonal $[M/M]$ rational approximation

$$\hat{\Phi}(v,z) \approx R_M(v,z) = \frac{P_M(z;v,T)}{Q_M(z;v,T)}, \qquad z = T^{\mu}.$$

After specializing $z = T^{\mu}$, the rational function $\Phi_M(v,T) = T\cdot z\cdot P_M/Q_M$ is rational in v at fixed T (Corollary 3.4, `cor:rational_Phi`), with $2M$ poles $\{u_j(T)\}$ in the complex v-plane.

These poles are **modes of the cumulant generating function as a function of the Fourier variable**, not quadrature nodes of any positive measure and not time-grid points. There is no time discretization anywhere — T enters continuously through $z = T^{\mu}$.

What is genuinely orthogonal-polynomial about the construction is the following. The Padé denominator Q_M is the orthogonal polynomial of degree M under the Hankel functional $\mathcal{L}[w^k] = M_k(v)$ (Theorem 3.1, `thm:three_term`), with the J-fraction expansion of $\hat{\Phi}$ in Theorem 3.6 being the standard Stieltjes continued fraction associated with this functional. But $\mathcal{L}$ is **not** a positive Hamburger/Stieltjes functional in the rough-Heston setting — the moments $M_k(v)$ are complex (polynomials in v with complex coefficients at fixed real v), the Hankel determinants H_n are merely assumed quasi-definite (nonzero), and the orthogonal-polynomial theory used is the **formal** one (Chihara, Brezinski) that does not require positivity.

Consequently:

- The poles $u_j(T)$ are **complex**, off the real line (Lemma 4.4 explicitly requires $u_j \notin \mathbb{R}$).
- The residues $A_j(T)$ are **complex**.
- The Hermitian symmetry $\phi_M(-\bar{v}, T) = \overline{\phi_M(v, T)}$ (Lemma 4.3, `lem:herm`) splits the poles into M in the upper half-plane and M in the lower — this is the reflection symmetry inherited from $\hat{\Phi}$ being a characteristic exponent, not a Gauss-quadrature property.

The right way to read the modal structure is therefore:

> The PFE $\Phi_M(u, T) = -\frac{1}{2}\sigma_T^2 u^2 - i\mu_T u + \sum_{j=1}^{2M} A_j(T)/(u - u_j(T))$ is a **modal expansion of the cumulant exponent in the Fourier variable**, with each pole $u_j(T)$ representing an exponential-decay mode $e^{-iu_j(T)x}$ of the density and $A_j(T)$ the modal amplitude. The order M sets how many modes are resolved; the limit $M \to \infty$ recovers the full continuous mode content of the exact $\hat{\Phi}$, off the Stahl branch/polar set.

Stahl convergence (Theorem 4.1, `thm:conv`) is then the standard statement that the diagonal Padé approximation of an analytic function with branch points converges in capacity on the complement of the (minimal-capacity) Stahl set — i.e., the modes resolved by order M become asymptotically dense in the continuous mode structure of the exact $\hat{\Phi}$.

The Gaussian real-axis decay of $|\phi_M(u, T)|$ (Lemma 4.4, `lem:decay`) is **not** a discrete-spectral-support phenomenon. It comes directly from the explicit Gaussian factor $e^{-\sigma_T^2 u^2/2}$ in the PFE (eq. `eq:PFE`), with the pole-sum contribution bounded uniformly by the constant . The Padé approximant has imposed a Gaussian leading-order shape on ϕ_M; the true ϕ has only stretched-exponential decay. Remark 4.6 (`rem:tail_compare`) is calling out that this Gaussian envelope is an **approximation artifact** in the tail — sharper than the truth at high $|u|$ — and is what makes COS pricing (Theorem 6.1) error decay at Gaussian rate against ϕ_M, even though it would not against the exact ϕ.

9. The M-Wright Density as the Atomic Bernstein Measure

We collect the role of the M-Wright function M_μ in the full picture:

- $M_\mu(r)\,dr$ is the Bernstein–Widder spectral measure of the Mittag-Leffler function $E_\mu(-t^\mu)$ (Pollard 1948).
- It is the density of the inverse μ-stable subordinator at time 1, i.e., the density of L_μ where $L_\mu := \inf\{s : \sigma_\mu(s) > 1\}$ and σ_μ is the standard μ-stable subordinator.
- It is a Fox H function: $M_\mu(r) = H^{1,0}_{1,1}\left[r \,\middle|\, \begin{matrix}(1-\mu,\mu)\\(0,1)\end{matrix}\right]$.
- It governs the time-fractional diffusion equation $D^\mu_C u = \partial^2_x u$: the fundamental solution at (x,t) is $\frac{1}{2t^{\mu/2}} M_{\mu/2}(|x|/t^{\mu/2})$ (Mainardi 1996).

In the rough Heston context, M_μ is the **atomic spectral density**: every more complex Bernstein–Widder spectral function that appears in the construction (the spectral density of $E_{\mu,\mu}(-\lambda t^\mu)$ in (6); the Prabhakar densities; the single-pole densities of Section 6 above; the multi-convolution densities of the subset-sum formula in Theorem 5.1) is obtained from M_μ by convolution, Mellin shift, or analytic continuation. M_μ is to the rough Heston density what the Gaussian is to the Black–Scholes density: the elementary spectral atom out of which everything is built.

10. The Logical Chain: A Summary Table

The construction in `frmp.tex` realizes the following chain of Bernstein–Widder identifications, layer by layer.

Object in `frmp.tex`	Bernstein–Widder role	Spectral measure
Kernel $t^{\mu-1}/\Gamma(\mu)$ of I^μ (eq.\ (1) above)	Completely monotone	$s^{-\mu}/\Gamma(1-\mu)\,ds$ — the μ-stable subordinator Lévy density
Linearized solution $E_\mu(-\lambda t^\mu)$ (Section 4 above)	Completely monotone (Pollard 1948)	$M_\mu(r)\,dr$ — the M-Wright density
Two-parameter solution $E_{\mu,\mu}(-\lambda t^\mu)$	Completely monotone (Schneider 1996)	$p_{\mu,\lambda}(s)\,ds$, explicit (eq.\ (6) above)
Prabhakar $E^\gamma_{\mu,\mu}(-\lambda t^\mu)$	Completely monotone (Mainardi–Garrappa 2014, with constraints)	Stieltjes-transform density, Fox-H representable
Padé denominator Q_M (Theorem 3.2)	Orthogonal polynomial under Hankel moment functional $\mathcal{L}$ (formal, complex-valued; not Stieltjes-positive)	Zeros = the $2M$ modes $u_j(T)$ of the Padé cumulant exponent in the Fourier v-plane
Pole factors $\exp(A_j/(u-u_j))$ in PFE of ϕ_M (eq.\ `eq:PFE`)	Fourier inverse is the Bernstein–Widder spectral density of a one-pole Mittag-Leffler factor	Single-Bessel Fox-H density (Section 6 above)
Density formula $f_M(x,T)$ (Theorem 5.1, eq.\ `eq:foxH_density`)	Multiple convolution of one-pole Bernstein–Widder densities against Gaussian	Finite sum of Fox-H functions over subsets S

Object in `frmp.tex`	Bernstein–Widder role	Spectral measure
Newton–Girard collapse (Corollary 5.1.1)	Symmetric-function reduction of 2^{2M} Bernstein–Widder convolution summands	$2M+1$ symmetric groups, FFT-evaluable
Price formula (Theorem 5.2, eq.\ `eq:foxH_price`)	One additional Mellin shift = one additional Gamma factor per Fox-H increment	(m,n,p,q) shifted by $(+1,+1,+1,+1)$

The unifying statement is: **every Mittag-Leffler-type object in the construction is completely monotone (under the parameter ranges relevant to rough Heston), and therefore admits, by Bernstein, a unique nonnegative spectral measure realized concretely as a Fox-H function via the Mellin–Barnes representation.** The paper's three theorems (T1: Stahl convergence; T2: Fox-H closed form; T3: COS alternative) are, respectively, (T1) the discrete approximation of the spectral measure converges off its support, (T2) the spectral measure can be written in closed form via Mellin–Barnes, and (T3) the same answer can be obtained by Fourier quadrature on the truncated spectral approximation.

11. Why This Matters Numerically

The Gaussian real-axis decay of ϕ_M (Lemma 4.4) is **strictly sharper** than the stretched-exponential decay $e^{-c|u|^{2\mu}}$ of the exact rough-Heston characteristic function (Remark 4.6, `rem:tail_compare`). The right reading is:

- The exact $\phi(u,T)$ has $|\log\phi(u,T)|$ growing like $|u|^{2\mu}$ at high frequency, reflecting the genuine fractional regularity $\mu<1$ of the underlying process.
- The Padé $\phi_M(u,T)$ has, by construction, an explicit Gaussian leading factor $e^{-\sigma_T^2u^2/2}$ extracted in the PFE (eq. `eq:PFE`), with the rest of the modal sum bounded by the finite constant . The high-frequency behavior of ϕ_M is therefore Gaussian-dominated, **not** stretched-exponential. This is a tail-behavior **mismatch** between ϕ_M and ϕ, confined to high $|u|$ outside the strip where the Padé moment-matching is operative.
- Convergence as $M\to\infty$ (Stahl, Theorem 4.1) is uniform on compacta in v off the Stahl set, not uniform globally — so the high-frequency mismatch is consistent with capacity-convergence and does not contradict it.

What the Padé buys numerically is: a representation in which the modal content responsible for the bulk and near-tail of the density is resolved exactly (through $2M$ modes), with a Gaussian envelope that makes COS pricing have Gaussian-rate truncation error (Theorem 6.1, `thm:price`, eq. `eq:cos_error`) and that makes the Fox-H Mellin–Barnes contours absolutely convergent.

The $O(M^2)$ front-end cost (Chebyshev algorithm), $O(M^2)$ per maturity (PFE), and $O(M\log^2 M)$ per strike (Fox-H via Newton–Girard FFT) decomposition of Section 7 is the computational image of:

- **Front-end $O(M^2)$**: orthogonal-polynomial recursion against the Hankel functional $\mathcal{L}$ on the Müntz moments — formal orthogonal-polynomial theory, not Gauss quadrature.

- **Per maturity** $O(M^2)$: root-finding of $Q_M(T^\mu; v) = 0$ in v — locating the $2M$ modes $\{u_j(T)\}$ of the Padé approximation in the v-plane.
- **Per strike** $O(M \log^2 M)$: Newton–Girard symmetric-function collapse of the subset sum in Theorem 5.1 — FFT polynomial multiplication of $\prod_j (1 + t x_j)$.

12. The Closed Circle

The five layers of the construction in `frmp.tex` are five constructive realizations of Bernstein–Widder applied to the rough-Heston characteristic exponent:

1. **The kernel** $t^{\mu-1}/\Gamma(\mu)$ of the fractional integral I^μ is completely monotone — this is what makes the Müntz lattice $\{t^{k\mu}\}$ close under I^μ and what makes the Müntz–Tau recurrence well-posed.
2. **The linear Mittag-Leffler solution** $E_\mu(-t^\mu)$ is completely monotone (Pollard) — this is what guarantees that the linearized fractional Riccati equation has a positive, well-behaved fundamental solution, which the Müntz–Tau series perturbs.
3. **The Hankel functional** $\mathcal{L}[w^k] = M_k(v)$ is quasi-definite (Hankel determinants nonzero), supporting a formal orthogonal-polynomial theory; the diagonal Padé approximation R_M converges in capacity off the Stahl branch/polar set (Stahl). The poles $u_j(T)$ are the modes of Φ_M in the Fourier v-plane — complex, off the real line, paired by Hermitian symmetry — and they are **not** quadrature nodes of any positive spectral measure.
4. **Each pole factor** $\exp(A_j/(u - u_j))$ of the Padé characteristic function is, on Fourier inversion, the Bernstein–Widder spectral density of a single-pole generalized Mittag-Leffler factor — i.e., a Fox H function of type $H^{1,0}_{0,2}$ (single Bessel).
5. **The full density** is the convolution of these single-pole Bernstein–Widder densities against the Gaussian envelope — by Bernstein–Widder closure under products (B1), this is again a Fox H, summed over subsets S, with parameter data $(|S| + 1, |S|, |S| + 1, 2|S| + 1)$ and amplitude data given by elementary symmetric functions of (A_j, u_j).

The Bernstein–Widder content of the pipeline lives at the level of the **individual modal factors** of ϕ_M, not at the level of Φ_M itself. Each one-pole exponential $\exp(A_j/(u - u_j))$ is, on Fourier inversion, a completely-monotone-type Mittag-Leffler / Bessel object that carries a continuous Bernstein–Widder spectral density of its own — and that density is named in closed form by a single-Bessel Fox H. The full density $f_M(x, T)$ is the convolution of these spectral densities against the Gaussian envelope, and Bernstein–Widder closure under products (B1) is the abstract reason the convolution stays in the Fox-H class. The Mellin–Barnes representation is the universal closed-form name for the resulting convolutions.

The paper is — through this lens — a theorem about how to take a rough-Heston cumulant exponent whose modal structure in the Fourier variable is implicit (a nonlinear fractional Volterra equation) and make that modal structure **algorithmically explicit** as a finite collection of poles $\{u_j(T), A_j(T)\}$, each of which, by Bernstein–Widder applied to its Fourier-inverted spectral density, becomes a Fox H. The Fox H-function is the universal closed form because

Mellin–Barnes is the universal language for the Bernstein–Widder spectral integrals of Mittag-Leffler-type factors.